BCS Critical Temperature in SCm Vacuum

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PAPER_950: BCS Critical Temperature in SCm Vacuum

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** bcs_superconductivity_uqff.py (BCSCriticalTemperature) **Calculator:** BCSCriticalTemperatureCalc (CP4 #534) **CVW:** v2.0.0 compliant

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Abstract

We compute the BCS critical temperature T_c in the SCm vacuum phonon framework. The relation $T_c = (1.13 \cdot \hbar\omega_{\text{SCm}}/k_B) \cdot \text{exp}(-1/N(0)V_{\text{SCm}})$ replaces the conventional Debye cutoff with $\omega_textSCm = 2\pi \times 1.25$ THz, yielding critical temperatures governed by the SCm phonon attraction strength V_{SCm} and Fermi-level density of states $N(0)$.

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1. Critical Temperature

$$T_c = \frac{1.13\hbar\omega_textSCm}{k_B}\,\text{exp!}\left(-\frac{1}{N(0)V_{\text{SCm}}}\right)$$

2. BCS Gap Relation

$$\Delta(0) = 1.764\,k_BT_c$$

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3. Source Data

- **File:** bcs_superconductivity_uqff.py
- **Session:** 214
- **CP4 Class:** BCSCriticalTemperatureCalc (#534)

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. Bardeen, J., Cooper, L.N. & Schrieffer, J.R. (1957) — Phys. Rev. 108, 1175
3. PAPER_949 — BCS Gap Equation in SCm Vacuum
4. PAPER_951 — Cooper Pair Phonon Coupling
5. PAPER_957 — Cooper Pair Lagrangian Variation

Cross-Links

Related Paper	Relationship
PAPER_949	Gap equation from which T_c is derived
PAPER_951	Cooper pair coupling strength V_{SCm}
PAPER_955	BCS gap at phonon resonance
PAPER_877	Cosmogenesis master Lagrangian

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	5.0×10^{-4} day^{-1}	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm phonon frequency	$\omega_textSCm$	$2\pi \times 1.25$ THz	Phonon resonance
SCm vacuum density	$\rho_textSCm$	7.09×10^{-37} kg/m^3	Fundamental

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SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
T_c formula	$T_c = 1.13\hbar\omega_{\text{SCm}}/k_B \cdot e^{-1/N(0)V_{\text{SCm}}}$	Mapped to SCm
BCS ratio $2\Delta_0$/k_BT_c	3.528	Standard BCS
SCm phonon replaces Debye	$\omega_D \to \omega_textSCm$	Novel

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Critical Temperature (BCS-SCm Thermal Threshold)

§A.2 Lagrangian Density

$$\mathcal{L}_{T\_c} = -N(0)|\Delta|^2/V_{\mathsf{SCm}} + k_B T \ln Z_{\mathsf{phonon}}(\omega_t ext SCm)$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\left.\frac{\partial \mathcal{L}}{\partial T}\right|_{T=T\_c} = 0 \implies T_c = \frac{1.13\hbar\omega_{\mathsf{SCm}}}{k_B} \,\text{exp!}\left(-\frac{1}{N(0)V_{\mathsf{SCm}}}\right)}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms → SCm vacuum → phonon $\omega_t ext SCm$ → BCS gap → T_c thermal threshold → condensate onset

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§B VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

$$\mathsf{VDS}(T) = \rho_t ext SCm \cdot (1 - (T/T_c)^4) \cdot S_{26}$$

§B.2 Dipole Vortex Primes (DVP)

The T_c threshold maps to prime $p = 2$ (Cooper pair symmetry).

§B.3 Buoyancy Saturation Harmonics (BSH)

$$\mathsf{BSH}(T) = \text{tanh!}\left(\frac{T_c - T}{T_c}\right) \cdot S_{26}$$

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
κ decay	5×10^{-4}	Confirmed
$[SSq]$	0.57	Confirmed